**Example:** Let $X = \{a, b, c, d\}$ &

$\mathcal{S} = \{\{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{c, d\}\}$. Find the topology $\tau$ generated by $\mathcal{S}$. Can you name $\tau$?

**Solution:**

$\mathcal{S}^{\cap} = \{\emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{c, d\}, X\}$

is a base for $\tau$.

$\tau = \{\emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}, \{a, b, c\}, \{a, c, d\}, \{a, b, d\}, \{b, c, d\}, X\}$

$\therefore$ $\tau$ is the discrete topology on $X$.

Now we reexpress some of the results we have considered earlier on in terms of subbases;

**Theorem:** Let $(X, \tau)$, $(Y, \vartheta)$ be topological spaces, $\mathcal{S}$ a subbase for $\vartheta$ & $f: X \to Y$ be a function. Then $f$ is $\tau$-$\vartheta$ continuous $\iff$ $f^{-1}(S) \in \tau$ for every $S \in \mathcal{S}$.

**Proof:**

$\Longrightarrow$ It is clear from the def$^{n}$ of continuous functions.

$\Longleftarrow$ **Step 1** Let $\mathcal{B}$ be an arbitrary base for the topology $\vartheta$ on $Y$. Suppose that $f^{-1}(B) \in \tau$ for every $B \in \mathcal{B}$. We shall show that $f$ is continuous;

Let $V \in \vartheta$. Then there exists a subclass $\mathcal{B}'$ of $\mathcal{B}$ s.t. $V = \bigcup \mathcal{B}'$

& $f^{-1}(V) = f^{-1}\left(\bigcup \mathcal{B}'\right)$

$= \bigcup f^{-1}(\mathcal{B}')$

$= \bigcup \{ f^{-1}(B) \mid B \in \mathcal{B}' \subseteq \mathcal{B} \} \in \mathcal{T}$

since $f^{-1}(B) \in \mathcal{T}$ for every $B \in \mathcal{B}$ & arbitrary union of open sets is open.

Step 2 Assume that $f^{-1}(S) \in \mathcal{T}$ for every $S \in \mathcal{S}$. We shall show that $f$ is continuous:

Because $\mathcal{S}$ is a subbase for $\mathcal{V}$, $\mathcal{B} = \mathcal{S}^{\cap}$ is a base for $\mathcal{V}$.

If $B \in \mathcal{B}$ then $B = \bigcap \mathcal{S}'$ where $\mathcal{S}'$ is a finite subclass of $\mathcal{S}$. Hence

$f^{-1}(B) = f^{-1}\left(\bigcap \mathcal{S}'\right) = \bigcap f^{-1}(\mathcal{S}')$

$= \bigcap \{ f^{-1}(S) \mid S \in \mathcal{S}' \} \in \mathcal{T}$

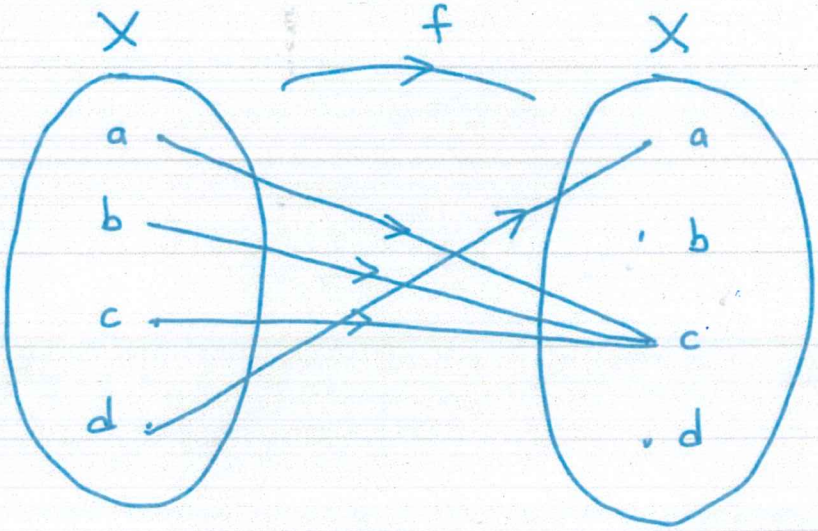
since $f^{-1}(S) \in \mathcal{T}$ for every $S \in \mathcal{S}$ & finite intersections of open sets is open.

$\therefore$ $f$ is continuous by Step 1.

Example: Let $X = \{a, b, c, d\}$,

$\mathcal{S} = \{\{a, b\}, \{a, c\}, \{d\}\}$. Then the topology $\mathcal{T}$ generated by $\mathcal{S}$ is given by

$\mathcal{T} = \{\emptyset, \{a\}, \{d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{a, b, d\}, \{a, c, d\}, \{a, b, c\}, X\}$. Let $f : X \longrightarrow X$ be the function defined by




Show that $f$ is continuous.

Sol^n

$f^{-1}(\{a,b\}) = \{d\} \in \tau$

$f^{-1}(\{a,c\}) = X \in \tau$

$f^{-1}(\{d\}) = \phi \in \tau$

i.e. $f^{-1}(S) \in \tau$ for every $S \in \mathcal{S}$.

$\therefore$ $f$ is continuous.

**Theorem:** Let $(X, \tau)$ be a topological space, $\mathcal{S}$ a subbase for $\tau$ & $(x_n)$ be a sequence of points in $(X, \tau)$. Then $\lim_{n \to \infty} x_n = a$, where $a \in X \iff$ there exists an integer $n_0 > 0$ s.t. $n \geq n_0 \Rightarrow x_n \in S$ for every $S \in \mathcal{S}$ with $a \in S$.

**Proof:**

$\Rightarrow$ It is clear from the definition of convergent sequences in topological spaces.

$\Leftarrow$ Assume that there exists an $n_0 \in \mathbb{Z}^+$ s.t. $n \geq n_0 \Rightarrow x_n \in S$ for every $S \in \mathcal{S}$ with $a \in S$. We shall show that $x_n \longrightarrow a$ (as $n \to \infty$):

Let $U \in \tau$ with $a \in U$. Since $\mathcal{S}$ is a subbase for $\tau$ there exist a finite number of sets $S_1, S_2, \ldots, S_m \in \mathcal{S}$ s.t. $a \in S_1 \cap S_2 \cap \ldots \cap S_m \subseteq U$.

$\Rightarrow \quad a \in S_i \quad$ for $\quad i = 1, 2, \ldots, m.$

It follows from the hypothesis that there exists an $n_i \in \mathbb{Z}^+$ s.t. $n \geq n_i \Rightarrow$ $x_n \in S_i \quad$ for $\quad i = 1, \ldots, m.$

Let $n_0 = \max \{ n_1, \ldots, n_m \}$. Then

$n \geq n_0 \Rightarrow n \geq n_i \Rightarrow x_n \in S_i$ for each $i$.

$\Rightarrow x_n \in S_1 \cap \ldots \cap S_m \subseteq U.$

$\therefore \quad \lim_{n \to \infty} x_n = a.$

Example: Let $(X, \mathcal{T})$ be a topological space given in the last example.

$( X = \{a, b, c, d\}, \mathcal{S} = \{ \{a, b\}, \{a, c\}, \{d\} \} )$

Do the following sequences converge:

$(x_n) = (c, c, c, \ldots\ldots)$

$(y_n) = (a, c, a, c, \ldots\ldots)$ ?

Soln

$x_n \not\to a$ ( Take $a \in \{a, b\} \in \mathcal{S}$. For $n \geq 1$, $x_n = c \notin \{a, b\}$ )

$x_n \not\to b$ ( Take $b \in \{a, b\} \in \mathcal{S}$. For $n \geq 1$

$x_n = c \notin \{a, b\}$ )

$x_n \to c$ ( There is only one $S \in \mathcal{S}$ with $c \in S$, namely $S = \{a, c\}$ & for $n \geq 1 = n_0$ $x_n = c \in S$. )

$x_n \not\to d$ ( $d \in \{d\} \in \mathcal{S}$. But $x_n = c \notin \{d\}$ for $n \geq 1$. )

$y_n \not\to a$ ( $a \in \{a, b\} \in \mathcal{S}$. But $y_2 = y_4 = \dots = y_{2n} = c \notin \{a, b\}$ )

$y_n \not\to b$ ( $b \in \{a, b\} \in \mathcal{S}$. But $y_2 = y_4 = \dots = y_{2n} = c \notin \{a, b\}$ )

$y_n \to c$ ( There is only one set namely $\{a, c\}$ in $\mathcal{S}$ containing $c$ & for all $n \geq 1$ $y_n \in \{a, c\}$ )

$y_n \not\to d$ ( $\{ d \in \{d\}$ but

$y_n \notin \{d\}$ for any $n \geq 1$ )